

Lab 09b – Implement Azure Container Instances (ACI)

What problem are we solving?

Your org:

- Has apps running on **VMs**
- Wants **no server management**
- Wants **fast deployment**
- Doesn't want **Kubernetes complexity**

 **Azure Container Instances** is the solution.

Core Concepts (Before Tasks)

What is a Container?

- Lightweight package of:
 - App code
 - Runtime
 - Dependencies
- Runs the same everywhere
- Faster than VMs

What is Azure Container Instance (ACI)?

- **Serverless containers**
- No VM, no OS management
- Pay **per second**
- Ideal for:
 - Simple web apps
 - Batch jobs
 - Testing & demos
 - Short-lived workloads

 Not ideal for:

- Long-running complex apps
 - Advanced networking
 - Orchestration → use AKS instead
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Task 1 – Deploy Azure Container Instance

What you are doing

You are:

- Pulling a **prebuilt Docker image**
- Running it directly in Azure
- Exposing it over **HTTP (port 80)**

Key Settings Explained

Setting	Why
Container name	Logical identifier
Image Source: Quickstart	Microsoft-maintained sample
Image	Simple Hello World web app
DNS Name Label	Public URL for the container
Linux image	Cheaper & faster

 Important:

<dns-name>.<region>.azurecontainer.io

This becomes your public website.

Why no Networking config?

ACI:

- Automatically exposes **port 80**
 - Creates public endpoint
 - No VNet needed for this lab
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Task 2 – Test and Verify Deployment

What you verify

1. **Container Status**
 - Must be **Running**
 2. **Public Access**
 - Open FQDN in browser
 3. **Application**
 - "Welcome to Azure Container Instance"
 4. **Logs**
 - HTTP GET requests appear
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Why logs matter?

- Proves:
 - App is serving traffic
 - Container is healthy
 - Requests are being handled

This replaces:

- VM event logs
 - Web server logs
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Cleanup – Why Important?

ACI is billed per:

- CPU
- Memory
- Seconds

Even idle containers **cost money**

-  Always delete the **resource group**
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BIG PICTURE – What You Learned

Concept	Explanation
Serverless containers	No VM or OS
Docker image	App packaged once
ACI	Fastest container deployment
Pay-per-second	Cost efficient
Logs	Built-in observability

Exam / Interview Ready Q&A

Q1: When should I use ACI?

Answer:

For simple, short-lived container workloads where infrastructure management is not required.

Q2: Difference between ACI and AKS?

ACI	AKS
No orchestration	Kubernetes orchestration
Single container	Multi-container apps
Very fast	More control
No scaling rules	Autoscaling

Q3: Does ACI need a VM?

Answer:

No. Azure manages the underlying compute automatically.

Q4: Can ACI be private?

Answer:

Yes, using VNets (not covered in this lab).

Q5: Is ACI PaaS or IaaS?

Answer:

Serverless container platform (closer to PaaS).